

WILLIAM ORTEGA

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EDUCATION

Harvey Mudd College, Claremont, CA

Expected May 2028

Bachelor of Science, Engineering

RELEVANT COURSEWORK: Fluid Mechanics, Continuum Mechanics, Chemical & Thermal Engineering Processes, Signals and Systems Engineering, Experimental Engineering, Differential Equations and Fourier Analysis, AI-Amplified Prototyping, Digital Electronics & Computer Engineering, Engineering Design, Multivariable Calculus, Probability & Statistics for Engineers

TECHNICAL SKILLS

- **Software:** Solidworks (CAD), MATLAB, Python, LabVIEW, COMSOL, Arduino/Teensy
- **Experimental:** Particle Image Velocimetry (PIV), high-speed imaging, data acquisition and processing, oscilloscope, function generator, multimeter
- **Fabrication:** soldering, mills, lathe, bandsaw, shear press, tablesaw, ShopBot
- **Languages:** English (Fluent), Spanish (Native)

RESEARCH EXPERIENCE

Undergraduate Research Assistant, Flow Imaging Lab — Harvey Mudd College

January 2026 - Present

- Characterize bio-inspired flapping-fin propulsion at the air–water interface; adapted thrust-processing code for horizontal configurations and compared thrust coefficients against vertical, surface-crossing cases
- Ran 100+ experimental trials sweeping actuation frequency (1–3 Hz), fin thickness (1/32–1/8 in), and five aspect ratios to map propulsive performance
- Conducted Particle Image Velocimetry (PIV) analysis — full optical setup, camera positioning, and laser-sheet alignment — to analyze wake flow fields and vortex structures; processed thrust and flow data in MATLAB
- Presenting horizontal-fin and aspect-ratio research at the APS Division of Fluid Dynamics Annual Meeting in Nov. 2026

ENGINEERING PROJECTS

Underwater ROV — Depth, Temperature & RGB Measurement System

January 2026 - May 2026

- Designed, assembled, and tested PVC frame, electronics, and motor system for submersible robot
- Designed and soldered custom PCB for real-time depth, temperature, and RGB light attenuation data acquisition
- Interfaced TCS34725 RGB color sensor and MPX5700 pressure sensor with a Teensy 4.1 microcontroller for synchronized depth-resolved spectral measurements

Mobile Lectern - Client Design Project (Team of 4)

September 2025 - November 2025

- Translated client needs into engineering specifications and led iterative concept generation, CAD modeling, prototyping, and fabrication for a portable lectern.
- Delivered a collapsible prototype under a \$50 budget with ~1.6 s setup time, >12 lb load capacity, and a complete SolidWorks drawing package.

Hammer - Machining Project

September 2025 - November 2025

- Fabricated hammer components from raw stock using milling, turning, and bandsaw operations; held drawing tolerances and validated hardface hardness requirements.

LEADERSHIP & ACTIVITIES

Founder & Chapter Co-President- SHPE Harvey Mudd College Chapter

January 2026 – Present

- Co-founded HMC's SHPE chapter and grew engagement to 52 students through recruiting, campus outreach, and professional-development programming.
- Secured \$10K in first-year funding (Engineering Dept, ASHMC, Student Opportunity Fund) to sponsor 12 members for the national SHPE conference
- Planned 3 professional-development workshops and launched company outreach for a next-year sponsorship pipeline year

Club President - SALSA (Society of Professional Latinx in STEM) Mudd

January 2025 – May 2026

- Led weekly meetings sharing academic and professional development opportunities with Latinx students
- Planned and executed 5+ cultural and social events per semester, including study sessions, carne asadas, and cultural movie nights
- Expanded cross-campus collaboration with five 5C Latinx organizations to strengthen community retention and belonging

EXPERIENCE

Orientation Mentor - Harvey Mudd Office of Residential Life

May 2026 - Present

- Led week-long orientation programming for the incoming Class of 2030; responsible for 57 first-year students through move-in and transition
- Completed Mental Health Blue Dot training; host ongoing dorm events to build residential community

Latino Sponsor - Chicano-Latino Student Affairs — Claremont Colleges

August 2025-May 2026

- Hosted monthly one-on-one support sessions with 5 CLSA mentees to ensure academic and social success
- Collaborated with Sponsor cohort to organize community-building events (Colombian Festival, Día de los Muertos, skill workshops) to increase retention and belonging

Mentor - Office of Institutional Diversity, Summer Institute — Harvey Mudd College

August 2025-May 2026

- Completed Mental Health Blue Dot training
- Led week-long transitional programming for first-generation, low-income, and international students
- Conducted ongoing check-ins with mentees throughout the academic year to support academic success

Residential Counselor - Upward Bound — Cal Poly Pomona

June 2025-July 2025

- Tutored approximately 30 high school students daily in Calculus and Pre-Calculus, supporting mastery of core concepts
- Led daily team-building activities to foster cohort engagement and belonging
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